INNOVATIVE EVALUATION PRACTICES FOR BCSE308P – COMPUTER NETWORKS LAB

FALL SEMESTER 2026–27

WI-FI QUALITY MAPPING AND DEAD-ZONE DIAGNOSTICS

YASHITA KUSHWAH	24BCE2473
ISHITA DUGGAL	24BCE2502
YASHIKA KUSHWAH	24BDS0364

Submitted in partial fulfilment of the requirements for
B.Tech. in Computer Science and Engineering

School of Computer Science and Engineering
Vellore Institute of Technology



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ABSTRACT

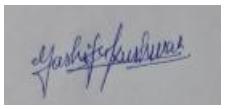
Wi-Fi Quality Mapping and Dead-Zone Diagnostics is a local Streamlit application that helps users understand how wireless signal strength changes across mapped locations. The project collects or accepts Wi-Fi readings, associates them with location IDs and X/Y coordinates in metres, validates the data, calculates location-level statistics, and displays the results through an interactive dashboard. Received Signal Strength Indicator (RSSI), measured in dBm, is used as the main signal-strength indicator; values closer to zero generally represent stronger received signal. The dashboard presents a coordinate-based coverage view, signal-quality categories, location comparisons, and an estimated coverage surface produced by inverse-distance-weighted interpolation. It also provides a configurable threshold for flagging locations whose mean RSSI is below the selected level. A baseline that flags a location when any individual reading is below the threshold can be compared with a mean-based approach that uses repeated readings to reduce sensitivity to a single low outlier. Supporting features include data exploration, manual measurement entry, optional Windows `netsh` network-detail prefill, CSV export, and downloadable text/HTML reports. The project was implemented with Python and Streamlit, using analysis and visualization libraries, and tested with four automated unit tests that passed. Example data discussed during development contained 28 readings across six location IDs; the dataset is small and intended to demonstrate the workflow, not to certify real-world coverage. RSSI alone cannot establish throughput, latency, packet loss, or loss of connectivity, and interpolated regions are estimates rather than extra measurements.

Keywords: Wi-Fi, RSSI, coverage mapping, interpolation, threshold-based diagnostics, Streamlit, data validation.

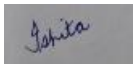
CERTIFICATE

This is to certify that the project report titled “WI-FI QUALITY MAPPING AND DEAD-ZONE DIAGNOSTICS” is a bonafide record of the work done by the following students in partial fulfilment of the requirements of the Innovative Evaluation Practices for BCSE308P – Computer Networks Lab during Fall Semester 2026–2027.

Yashita Kushwah (24BCE2473)

Signature: 

Ishita Duggal (24BCE2502)

Signature: 

Yashika Kushwah (24BDS0364)

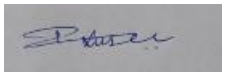
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1. INTRODUCTION

1.1 Background and motivation

Wi-Fi is widely used to connect laptops and mobile devices to local networks and the internet. Signal strength can vary from one point to another because of distance from the access point, walls, furniture, interference, antenna orientation, and changes in the environment. A single signal reading may not represent the usual condition at a location. Mapping repeated readings to physical coordinates makes it easier to compare locations and identify places that may need further investigation.

This project provides a simple, visual workflow for turning readings into location summaries and coverage diagnostics. It is designed for a lab demonstration: users can inspect measurements, compare locations, adjust a threshold, and explain how the system makes its flags.

1.2 Problem statement

Raw Wi-Fi readings are difficult to interpret when they are stored only as rows in a CSV file. Users need a consistent way to associate readings with positions, summarize repeated samples, visualize spatial patterns, and distinguish a genuinely persistent weak signal from an isolated low reading. The project addresses this by combining validation, descriptive statistics, mapping, and threshold-based comparisons in one dashboard.

1.3 Objectives

- Collect or enter Wi-Fi readings with timestamps, location IDs, coordinates and available network details.
- Validate records and normalize the input data before spatial analysis.
- Calculate mean, minimum, maximum and standard deviation of RSSI by location.
- Display measured locations and signal-quality categories on interactive visualizations.
- Estimate the signal pattern between measured points using inverse-distance weighting, while clearly labeling it as an estimate.
- Allow users to change the weak-signal threshold and compare any-reading and mean-based flags.
- Provide data export and downloadable analysis reports.

1.4 Scope and limitations

The project focuses on a local, small-scale mapping workflow. It does not control access points or perform a full wireless site survey. RSSI is not a direct measurement of internet speed, packet loss, latency, or user experience. A threshold flag is a prompt for investigation, not proof that connectivity is absent. Interpolation is approximate and can be misleading when there are too few points, points are unevenly distributed, or barriers are not represented. The optional Windows scan uses `netsh`, which generally exposes signal percentage and network details rather than raw RSSI; any conversion to dBm must be treated as an estimate.

1.5 Mapping to course modules and the lab experiment

The project applies computer-networking concepts related to wireless access, signal measurements, network performance observation, and empirical evaluation. It demonstrates how measurements from a network environment can be collected, organized, validated, summarized, and interpreted. It also follows an experimental approach by defining a threshold, comparing decision rules, and discussing the limitations of the observed data.

2. LITERATURE SURVEY

2.1 Overview of the existing protocol or scheme

IEEE 802.11 defines the family of wireless local area networking standards. In practical deployments, a device can observe signal-related information, but RSSI readings can vary with hardware, location, orientation, and environmental conditions. A basic monitoring approach may inspect a single reading or display a raw list. This project adds repeated-sample summaries and a spatial view to make comparisons easier.

2.2 Review of related work

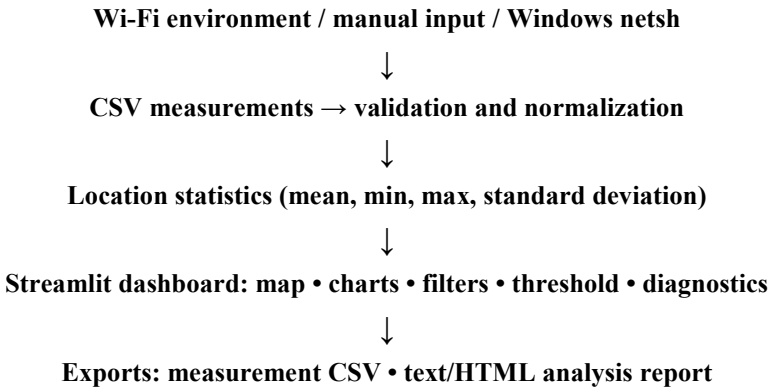
S. No.	Author / year	Technique / source	Metrics / tool	Merits	Gaps relevant here
1	IEEE 802.11 working group; standard family	Wireless LAN MAC/PHY specifications	Wi-Fi link and radio operation	Defines the core WLAN technology used in the environment	Does not itself provide this project's dashboard or threshold-comparison workflow
2	Bahl and Padmanabhan, 2000	RADAR: RF-based indoor location and tracking	Received signal measurements and location estimation	Demonstrates the value of RF measurements for indoor spatial reasoning	A full fingerprinting/location system is beyond this project's scope
3	Shepard, 1968	Inverse-distance-style interpolation for irregularly spaced data	Spatial interpolation	Provides a foundational approach for estimating values between samples	Interpolated values depend on point density and are not direct observations
4	Streamlit documentation; current framework	Python-based interactive data apps	Streamlit widgets and charts	Rapidly combines controls, data tables and visualizations	Dashboard correctness still depends on validation, tests and interpretation

2.3 Research gap and justification of novelty

The project does not claim a new wireless protocol or a new general-purpose interpolation method. Its project-level contribution is the integration of measurement validation, repeated-reading statistics, an interactive coordinate-based coverage view, a configurable threshold, a comparison between any-reading and mean-based decisions, and report export in a single beginner-friendly dashboard. The comparison helps demonstrate how the chosen decision rule affects the number of locations flagged.

3. PROPOSED SYSTEM

3.1 System architecture or topology



The dashboard uses the same validated measurement data for its summaries and visualizations. The spatial surface is computed from observed points and is explicitly treated as an estimate.

3.2 Baseline mechanism description

The baseline method marks a location if at least one reading is below the selected RSSI threshold. It is sensitive to short-lived drops and isolated outliers. For a location with readings r1, r2, …, rn and threshold T, the baseline flag is true if min(ri) < T.

3.3 Proposed mechanism: algorithm and pseudocode

The mean-based method first calculates the average RSSI for each location and marks it when that average is below T. This reduces sensitivity to one unusually low reading but can hide intermittent drops. Both views are useful and answer slightly different questions.

1. Read measurement records and normalize expected columns.
2. Reject or exclude rows missing essential location, coordinate, or valid RSSI values.
3. Group valid readings by location ID.
4. For each location, calculate mean, minimum, maximum, standard deviation and count.
5. Read the user-selected threshold T.
6. Baseline flag = any reading at the location is below T.
7. Mean-based flag = location mean is below T.
8. Display flags, charts and explanations; export report if requested.

3.4 Novelty statement

The novelty is an integrated educational workflow and live demonstration of how a threshold and an aggregation method change weak-coverage classification. The user can adjust T in the dashboard and observe the diagnostic results without editing code. This is an application-level contribution rather than a claim of a new protocol.

3.5 Assumptions and design decisions

- X and Y coordinates are supplied in metres relative to a chosen local origin; they are not GPS coordinates unless separately defined.
- Multiple readings are taken at each location to reduce reliance on one sample.
- RSSI values are negative dBm values; values closer to zero are generally stronger.
- A location is flagged by the mean-based rule only when mean RSSI is strictly below the threshold.
- Signal-quality categories are heuristic and are used for interpretation, not certification.
- Interpolated values between points are estimates and must not be counted as new measurements.

4. IMPLEMENTATION

4.1 Tools, platform, versions and hardware

Component	Use in project
Python 3.12.x (tested environment)	Core language and data processing
Streamlit	Local interactive dashboard and controls
Pandas / NumPy	Tabular cleaning, aggregation and numerical operations
Plotly / visualization libraries	Interactive charts and coverage views
pytest	Automated unit tests
VS Code and PowerShell	Development and execution environment
Windows laptop and Wi-Fi adapter	Local dashboard execution and optional `netsh` scan

Exact package versions can vary by environment. The repository's requirements.txt is the source of required Python packages. The optional live scan is Windows-specific; CSV-based analysis and dashboard functions can run on other platforms if dependencies are installed.

4.2 Simulation or testbed setup

This is a measurement-based testbed rather than a packet-level network simulator. Each location is assigned a location ID and local X/Y coordinates. The collector or manual form records available network details, and repeated readings are saved in CSV format.

Parameter	Project setting / example
Input file	data/wifi_measurements.csv
Columns	timestamp, location_id, x_m, y_m, ssid, rssi_dbm, signal_percent, band, channel, receive_rate_mbps, transmit_rate_mbps
Example dataset discussed	28 readings across 6 location IDs (P1–P5 and TEST1)
Default threshold	−80 dBm
Example threshold for method comparison	−55 dBm (illustrative, to show rule differences)
Coordinate unit	Metres in a local reference frame
Spatial estimate	Inverse-distance-weighted interpolation
Automated tests	4 tests in tests/test_analysis_core.py

4.3 Code structure and explanation of key modules

File / module	Responsibility
app.py	Main Streamlit dashboard, tabs, filters, threshold slider, KPI cards, diagnostics, measurement entry, exports and methodology explanations.
analysis_core.py	Reusable data cleaning, summary calculations, signal classification and interpolation preconditions.
wifi_collector.py	Command-line utility for recording Wi-Fi measurements.
validate_data.py	Checks input data quality, required values, duplicates and coordinate consistency.
analyze_data.py	Calculates summary statistics by location and writes a summary CSV.
visualize_wifi.py	Creates coverage maps and charts from measurements and summaries.
detect_dead_zones.py	Compares threshold-based baseline and mean-based potential dead-zone decisions.
generate_report.py	Generates a supporting analysis report.
tests/test_analysis_core.py	Automated tests for core analysis functions.

4.4 Challenges faced and how they were solved

- Data may contain missing or inconsistent fields; normalization and validation help prevent invalid rows from entering spatial analysis.
- Repeated readings can fluctuate; mean, minimum, maximum and standard deviation provide a more complete picture than one sample.
- The any-reading rule can flag a location due to one isolated drop; comparing it with mean-based classification shows this trade-off.
- Interpolation can be mistaken for measured data; dashboard explanations label the surface as estimated.
- Windows `netsh` does not provide raw RSSI in the same way as a direct measurement; the project treats any conversion as approximate.
- A local Streamlit app requires dependencies and the correct working directory; a requirements file and execution instructions document setup.

5. Experimental Setup and Performance Evaluation

5.1 Performance metrics and why they were chosen

Metric	Purpose
Mean RSSI (dBm)	Represents the average signal strength across repeated readings.
Minimum / maximum RSSI	Shows the weakest and strongest observed sample at a location.
Standard deviation (dB)	Shows spread/variability among repeated readings.
Reading count	Indicates how many samples support each location summary.
Locations flagged	Shows how the decision rule and threshold affect diagnostics.
Runtime / test pass count	Confirms basic automated analysis behavior; four unit tests passed.

5.2 Scenarios and parameter values

The dashboard supports changing the threshold interactively. The default is −80 dBm. To illustrate the difference between the two decision rules, a threshold of −55 dBm can be selected for the example readings below. This is an explanatory comparison, not a claim that −55 dBm is the best threshold for every environment.

5.3 Results: location summaries and baseline versus proposed

Location	n	Mean RSSI (dBm)	Sample SD (dB)	Min	Max
P1	5	-55.00	1.41	-57	-53
P2	5	-61.00	2.12	-64	-59
P3	5	-46.20	2.28	-49	-44
P4	5	-54.40	0.89	-56	-54
P5	5	-53.00	2.65	-56	-49
TEST1	3	-47.67	0.58	-48	-47

These descriptive values are calculated from the example readings listed for P1–P5 and TEST1 during project development. Confirm that the final CSV in the submitted repository contains the same readings before treating this table as the final dataset result.

Scenario / threshold	Baseline: any reading below T	Mean-based: mean below T	Interpretation
Default T = −80 dBm	0 of 6 locations	0 of 6 locations	No example location crosses this relatively permissive threshold.
Illustrative T = −55 dBm	4 of 6 locations (P1, P2, P4, P5)	1 of 6 locations (P2)	The any-reading rule responds to low individual samples; the mean rule flags only the location whose average is below the threshold.

Improvement percentage is not reported as a universal performance gain because the two methods classify different conditions and there is no independent ground-truth label for actual connectivity failure. The meaningful observation is the difference in flags at a specified threshold.

5.4 Analysis and discussion

P3 has the strongest average signal in this example (−46.20 dBm), while P2 has the weakest average (−61.00 dBm). RSSI closer to zero indicates stronger received signal, so P3's readings are stronger than P2's. At the default threshold of −80 dBm, none of these example means falls below the threshold. When the threshold is moved to −55 dBm, the baseline flags four locations because each has at least one sample below −55 dBm, while the mean-based rule flags only P2 because its average is below −55 dBm. This demonstrates the trade-off: the baseline can reveal intermittent drops but can be sensitive to one low sample; the mean-based rule is more stable but may hide occasional drops.

The map and interpolated surface are useful for visual comparison, but the surface depends on the number and distribution of measured points. Adding more positions and repeating measurements under consistent conditions would improve the reliability of the spatial picture.

5.5 Critical evaluation: strengths, weaknesses and failure conditions

Strengths	Weaknesses / failure conditions
One interface for data inspection, visualization, diagnostics and reporting.	Small datasets may not represent the whole building or changing network conditions.
Threshold can be adjusted live for a clear demonstration.	Threshold choice is environment-dependent and can change the number of flags substantially.
Repeated-sample statistics make variation visible.	Mean-based decisions can miss short intermittent drops.
Interpolation makes spatial trends easier to see.	Sparse or clustered points and physical barriers can make interpolated regions misleading.
Unit tests provide basic checks for analysis functions.	Four passing tests do not prove every dashboard path or live Wi-Fi scan is error-free.

6. Conclusion and Future Work

6.1 Summary of work and key findings

The project delivers a local dashboard that turns Wi-Fi measurements into location-level statistics, interactive coverage views and configurable weak-signal diagnostics. The example dataset discussed contains 28 readings across six location IDs, and four automated tests passed in 0.71 seconds. In the sample location readings, P3 has the strongest mean RSSI (−46.20 dBm) and P2 the weakest (−61.00 dBm). The comparison at −55 dBm shows that the any-reading baseline flags four locations while the mean-based method flags one. This is a classification difference, not a measured percentage improvement in real-world network performance.

6.2 Limitations

- Example data is small and may not match the final repository file exactly; verify before submission.
- RSSI alone does not measure throughput, latency, packet loss or prove that connectivity is unavailable.
- Thresholds are heuristic and require calibration for the environment.
- Interpolation is an estimate, not an additional measurement.
- The optional Windows scan is platform-specific and may provide signal percentage rather than raw RSSI.
- The current automated suite has four tests; broader integration and real-world validation are still needed.

6.3 Future enhancements

- Collect repeated readings at more points, over multiple times of day, and under recorded environmental conditions.
- Add ground-truth checks such as throughput, latency and packet-loss tests to validate whether RSSI flags correspond to user-visible problems.
- Add exportable charts and a downloadable raw-data validation report.
- Support floor-plan overlays and explicit wall/obstruction information.
- Expand automated tests to cover dashboard interactions, malformed uploads, report generation and threshold edge cases.
- Compare thresholds and aggregation methods across repeated trials and report mean ± standard deviation for the evaluation metrics.

7. REFERENCES

[1] IEEE, “IEEE Standard for Information Technology—Telecommunications and Information Exchange between Systems Local and Metropolitan Area Networks—Specific Requirements—Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications,” IEEE 802.11 standard family.

[2] P. Bahl and V. N. Padmanabhan, “RADAR: An In-Building RF-Based User Location and Tracking System,” in Proceedings of IEEE INFOCOM, 2000.

[3] D. Shepard, “A Two-Dimensional Interpolation Function for Irregularly-Spaced Data,” in Proceedings of the 1968 ACM National Conference, 1968, pp. 517–524.

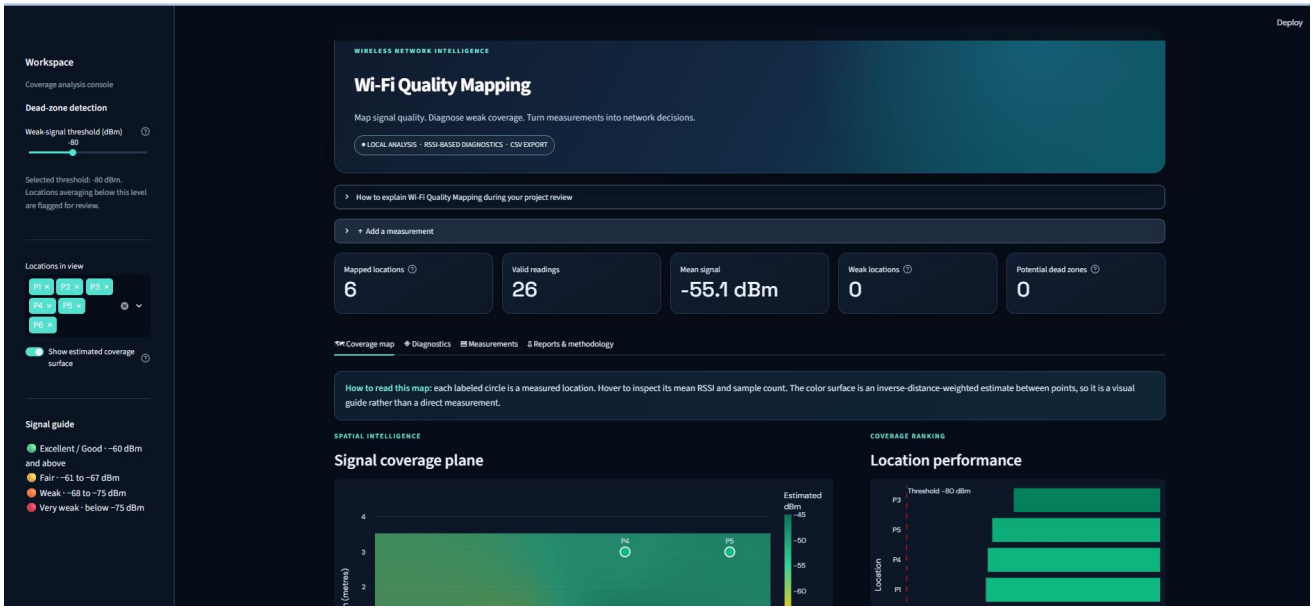
APPENDIX A: IMPORTANT ALGORITHM LISTING

Threshold comparison pseudocode:

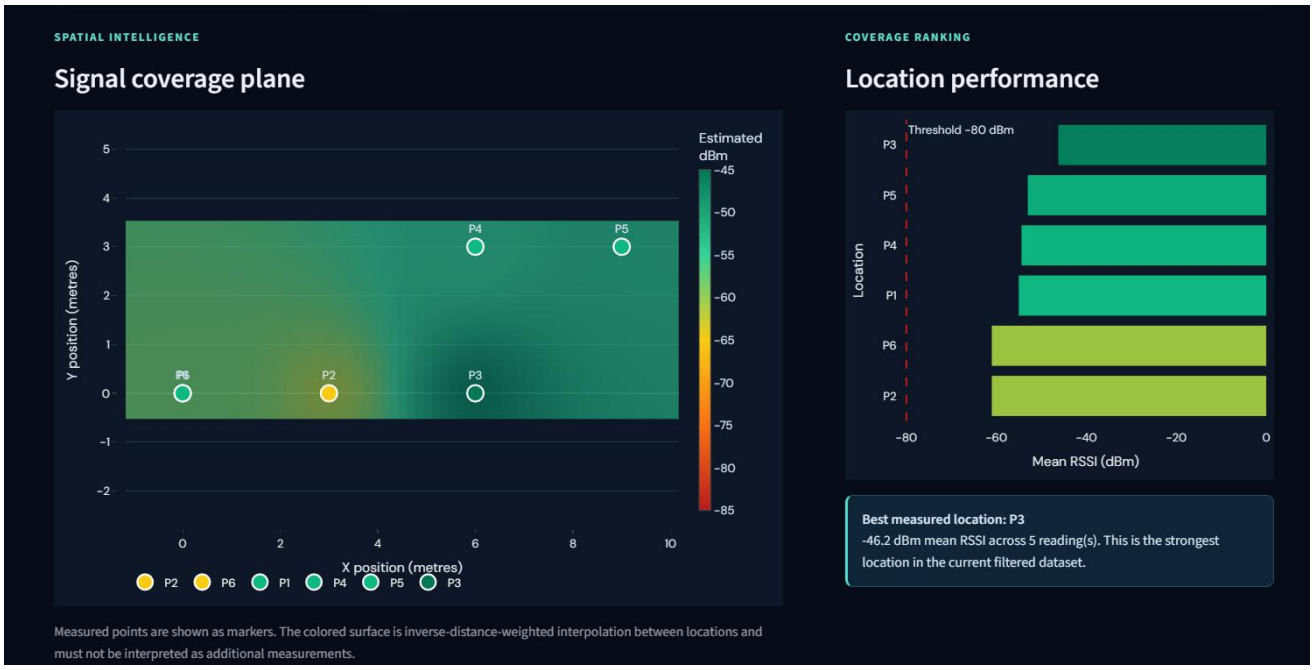
```
INPUT: readings grouped by location; threshold T
FOR each location L:
    values ← valid RSSI readings at L
    meanRSSI ← average(values)
    baselineFlag ← any(value < T for value in values)
    meanFlag ← (meanRSSI < T)
    display statistics and both flags
END FOR
OUTPUT: diagnostic table and report
```

APPENDIX B: SCREENSHOTS

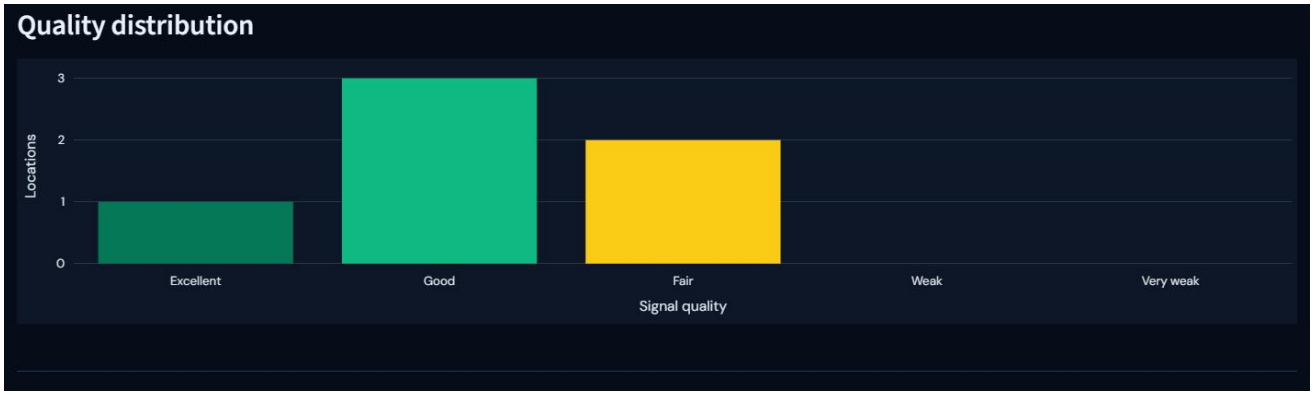
(1) dashboard overview



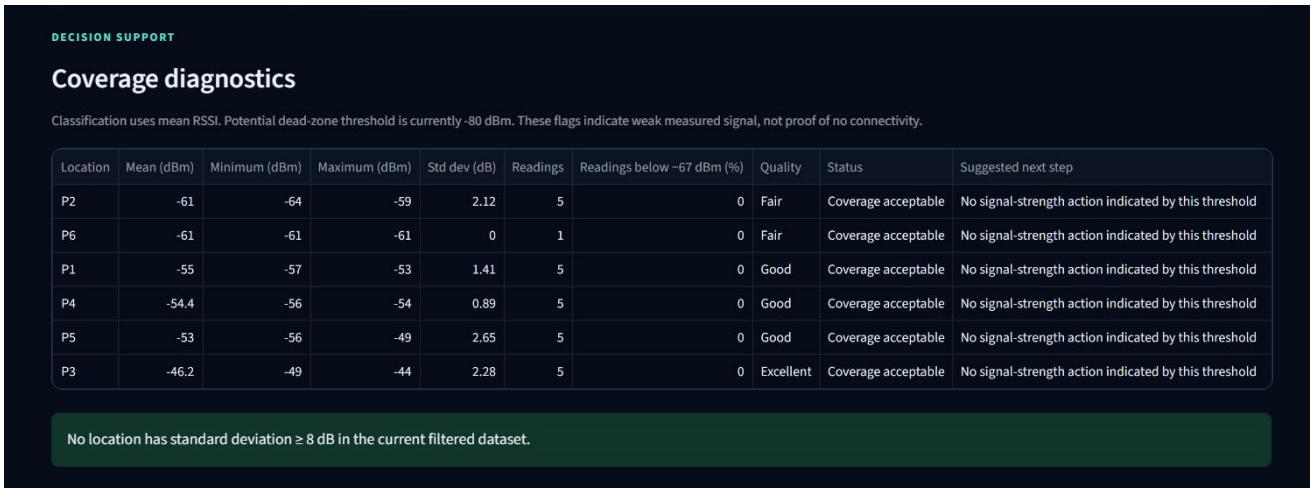
(2) coverage map



(3) quality distribution



(4) diagnostics



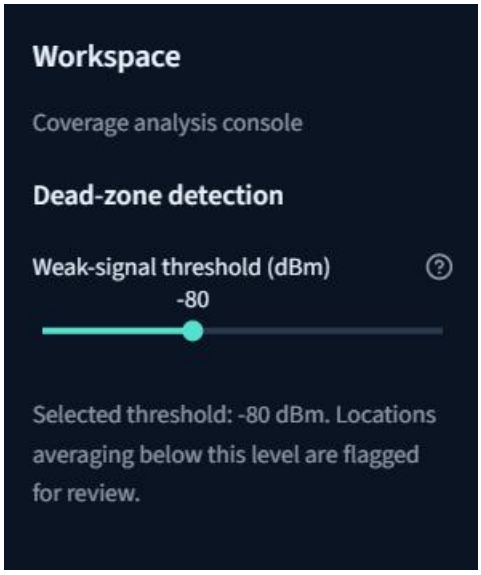
Location	Mean (dBm)	Minimum (dBm)	Maximum (dBm)	Std dev (dB)	Readings	Readings below −67 dBm (%)	Quality	Status	Suggested next step
P2	-61	-64	-59	2.12	5	0	Fair	Coverage acceptable	No signal-strength action indicated by this threshold
P6	-61	-61	-61	0	1	0	Fair	Coverage acceptable	No signal-strength action indicated by this threshold
P1	-55	-57	-53	1.41	5	0	Good	Coverage acceptable	No signal-strength action indicated by this threshold
P4	-54.4	-56	-54	0.89	5	0	Good	Coverage acceptable	No signal-strength action indicated by this threshold
P5	-53	-56	-49	2.65	5	0	Good	Coverage acceptable	No signal-strength action indicated by this threshold
P3	-46.2	-49	-44	2.28	5	0	Excellent	Coverage acceptable	No signal-strength action indicated by this threshold

Method comparison

The single-reading baseline flags a location if any reading falls below the threshold. The mean-based method flags it only when the average is below the threshold. The mean-based method is less sensitive to isolated outliers, but can hide intermittent drops; both views are useful.

Location	Mean RSSI	Minimum RSSI	Samples	Any reading below threshold	Mean below threshold
P1	-55	-57	5	☐	☐
P2	-61	-64	5	☐	☐
P3	-46.2	-49	5	☐	☐
P4	-54.4	-56	5	☐	☐
P5	-53	-56	5	☐	☐
P6	-61	-61	1	☐	☐

(5) threshold slider



(6) measurements

Measurement ledger

Filter by location ID or SSID

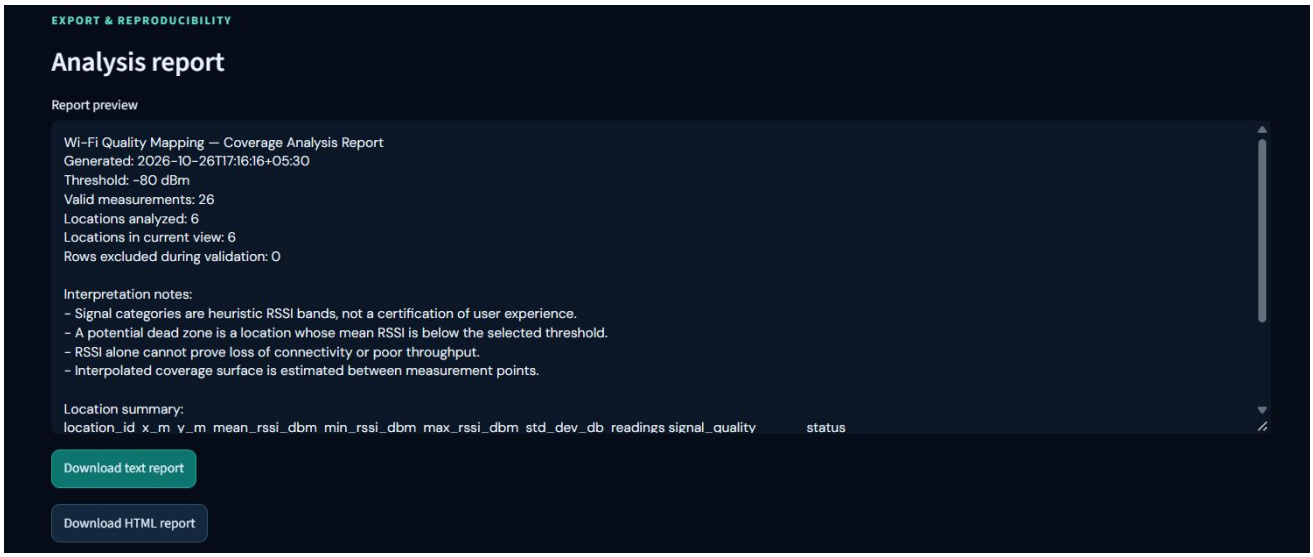
Type a location or network name

Showing 26 of 26 valid rows · 0 row(s) excluded by validation

timestamp	location_id	x_m	y_m	ssid	rssi_dbm	signal_percent	band	channel	receive_rate_mbps	transmit_rate_mbps
2026-10-16T21:32:20+05:30	P1	0	0	LHC-VIT	-53	100	2.4 GHz	13	144.4	144.4
2026-10-16T21:32:22+05:30	P1	0	0	LHC-VIT	-55	100	2.4 GHz	13	144.4	144.4
2026-10-16T21:32:24+05:30	P1	0	0	LHC-VIT	-55	100	2.4 GHz	13	144.4	144.4
2026-10-16T21:32:26+05:30	P1	0	0	LHC-VIT	-57	100	2.4 GHz	13	144.4	144.4
2026-10-16T21:32:29+05:30	P1	0	0	LHC-VIT	-55	100	2.4 GHz	13	144.4	144.4
2026-10-16T21:39:30+05:30	P2	3	0	LHC-VIT	-62	100	5 GHz	40	780	780
2026-10-16T21:39:32+05:30	P2	3	0	LHC-VIT	-59	100	5 GHz	40	866.7	866.7
2026-10-16T21:39:34+05:30	P2	3	0	LHC-VIT	-59	100	5 GHz	40	866.7	866.7
2026-10-16T21:39:36+05:30	P2	3	0	LHC-VIT	-64	100	5 GHz	40	866.7	866.7
2026-10-16T21:39:38+05:30	P2	3	0	LHC-VIT	-61	100	5 GHz	40	780	780

Download filtered measurements (.csv)

(7) report


EXPORT & REPRODUCIBILITY
Analysis report
Report preview
Wi-Fi Quality Mapping — Coverage Analysis Report
Generated: 2026-10-26T17:16:16+05:30
Threshold: -80 dBm
Valid measurements: 26
Locations analyzed: 6
Locations in current view: 6
Rows excluded during validation: 0
Interpretation notes:
- Signal categories are heuristic RSSI bands, not a certification of user experience.
- A potential dead zone is a location whose mean RSSI is below the selected threshold.
- RSSI alone cannot prove loss of connectivity or poor throughput.
- Interpolated coverage surface is estimated between measurement points.
Location summary:
location_id x_m y_m mean_rssi_dbm min_rssi_dbm max_rssi_dbm std_dev_db readings signal_quality status
Download text report
Download HTML report


Interpretation notes:
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- A potential dead zone is a location whose mean RSSI is below the selected threshold.
- RSSI alone cannot prove loss of connectivity or poor throughput.
- Interpolated coverage surface is estimated between measurement points.
Location summary:
location_id x_m y_m mean_rssi_dbm min_rssi_dbm max_rssi_dbm std_dev_db readings signal_quality status
P2 3.0 0.0 -61.0 -64 -59 2.12 5 Fair Coverage acceptable
P6 0.0 0.0 -61.0 -61 -61 0.00 1 Fair Coverage acceptable
P1 0.0 0.0 -55.0 -57 -53 1.41 5 Good Coverage acceptable
P4 6.0 3.0 -54.4 -56 -54 0.89 5 Good Coverage acceptable
P5 9.0 3.0 -53.0 -56 -49 2.65 5 Good Coverage acceptable
P3 6.0 0.0 -46.2 -49 -44 2.28 5 Excellent Coverage acceptable
Download text report
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